

CERTIFIED ETHICAL HACKER

CEH v13

🔧 POWERED BY AI

Think Like an Attacker.
Defend Like a Cybersecurity Professional.

Learn how attackers discover vulnerabilities and compromise systems, applications, networks and cloud environments — and how organizations detect, prevent and defend against those attacks.

20

Modules

221

Hands-On Labs

550+

Attack Techniques

4,000+

Hacking & Security Tools

01 THE PROGRAMME

What is CEH v13?

Certified Ethical Hacker (CEH) v13 is EC-Council's ethical hacking and cybersecurity certification program. It teaches participants how attackers identify, exploit and compromise vulnerabilities — and how organizations detect, prevent and defend against those attacks. CEH v13 integrates **AI-assisted ethical hacking** across the learning journey.

The ethical hacker's methodology

FROM ATTACKER MINDSET → DEFENSE



COURSE OBJECTIVE

See an attack from an ethical hacker's perspective

Develop the knowledge and practical skills to collect intelligence, discover systems, identify weaknesses, understand exploitation, test major attack surfaces and apply appropriate countermeasures.

CORE IDEA

Learn how attackers think and operate so you can identify weaknesses before malicious attackers exploit them.

WHO SHOULD JOIN?

Designed for technical professionals, career-changers and teams



IT professionals



Network engineers & administrators



System administrators



SOC / security analysts



Security administrators



IT auditors



Cloud & infrastructure professionals



IT support professionals



Graduates pursuing cybersecurity



Professionals transitioning into cybersecurity



People interested in ethical hacking / penetration testing



Organizations upskilling technical teams

02 BEFORE YOU START

New to Cybersecurity?

Here's what you should know before CEH.

CEH is not a basic computer-literacy course.

A participant can enter through the official training route without already being a cybersecurity professional — but a solid IT and networking foundation makes the intensive CEH learning experience considerably easier.



01 Networking FUNDAMENTALS

IP addressing TCP/IP
Ports & protocols DNS
DHCP Routers Switches
Firewalls LAN/WAN
HTTP/HTTPS



02 Operating Systems FUNDAMENTALS

Windows Linux basics
Files & directories
Users & permissions
Processes & services
Basic command-line usage



03 Security FUNDAMENTALS

Malware Authentication
Access control
Vulnerabilities Encryption
Firewalls
Common cyber threats

RECOMMENDED PATHWAY FOR A COMPLETE BEGINNER



**Complete
Beginner**
STEP 01



IT Fundamentals
STEP 02



**Networking
Fundamentals**
STEP 03



**Cybersecurity
Fundamentals**
STEP 04



CEH v13
STEP 05

KEY INSIGHT

Why does networking matter for ethical hacking?

Ethical hacking depends on understanding how computers, servers, ports, protocols and networks communicate. Networking knowledge at roughly a foundational / Network+ level can make **scanning, enumeration, traffic analysis and attack methodology** much easier to understand.

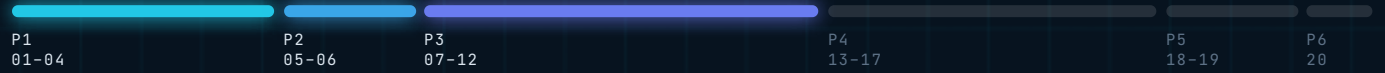


You do not need to be an expert hacker before CEH. The purpose of the course is to develop ethical hacking knowledge and skills. The stronger your fundamentals, the more value you can extract from the labs.

03 THE LEARNING JOURNEY

The 20-Module CEH Journey

The 20-module journey moves from foundations and discovery into exploitation concepts, specialized attack surfaces and defensive countermeasures. AI-assisted ethical hacking is integrated across appropriate areas of CEH v13.



PHASE 01

DISCOVER

Modules 01-04

01 Introduction to Ethical Hacking

Information security, cyber threats, ethical hacking methodologies, laws, standards and security controls.

02 Footprinting & Reconnaissance

OSINT, search engines, websites, email, WHOIS, DNS, social networks and reconnaissance.

03 Scanning Networks

Host discovery, ports, services, operating systems and network scanning.

04 Enumeration

Users, groups, resources, services, DNS and protocol enumeration.

PHASE 02

FIND THE WEAKNESS

Modules 05-06

05 Vulnerability Analysis

Identify, classify, assess and prioritize vulnerabilities.

06 System Hacking

Authentication attacks, gaining access, privilege escalation, maintaining access and attack evidence.

PHASE 03

UNDERSTAND ATTACKS

Modules 07-12

07 Malware Threats

Trojans, viruses, worms, ransomware, rootkits, fileless malware and malicious payloads.

08 Sniffing

Packet capture, traffic analysis, sniffing attacks and countermeasures.

09 Social Engineering

Phishing, impersonation, manipulation and credential-targeting attacks.

10 Denial-of-Service

DoS/DDoS attacks, detection and mitigation.

11 Session Hijacking

Session attacks, session-management weaknesses and countermeasures.

12 Evading IDS, Firewalls & Honeypots

Security monitoring, prevention technologies, evasion concepts and defensive countermeasures.

03 THE LEARNING JOURNEY · CONTINUED

From attack surfaces to protecting information

P1
01-04

P2
05-06

P3
07-12

P4
13-17

P5
18-19

P6
20

PHASE 04

ATTACK SURFACES

Modules 13-17

13 Hacking Web Servers

Web-server vulnerabilities, attacks, assessment and hardening.



14 Hacking Web Applications

Web-application vulnerabilities, attack methodologies and protection.



15 SQL Injection

SQL injection concepts, methodologies, detection and countermeasures.



16 Hacking Wireless Networks

Wi-Fi, authentication, encryption, wireless attacks and defensive controls.



17 Hacking Mobile Platforms

Mobile vulnerabilities, attack vectors and security controls.



PHASE 05

MODERN INFRASTRUCTURE

Modules 18-19

18 IoT & OT Hacking

IoT devices, Operational Technology, industrial environments and security threats.



19 Cloud Computing

AWS, Azure, Google Cloud, IAM, containers, Docker, Kubernetes, serverless, cloud vulnerabilities and controls.



PHASE 06

PROTECT INFORMATION

Modules 20

20 Cryptography

Encryption, hashing, symmetric/asymmetric cryptography, PKI, certificates, signatures and cryptanalysis.



FOUNDATIONS

Foundations & discovery

→ EXPLOITATION

Exploitation concepts

→ SURFACES

Specialized attack surfaces

→ DEFENSE

Defensive countermeasures



All 20 modules. AI-assisted ethical hacking is integrated across appropriate areas of CEH v13 — not confined to a single module.

04 HANDS-ON LAB EXPERIENCE

DON'T JUST LEARN HACKING. PRACTICE IT.

CEH is not purely theory-based. The official environment emphasizes practical exercises inside a controlled cyber range designed for authorized security practice.

221

HANDS-ON LABS

550+

ATTACK TECHNIQUES

4,000+

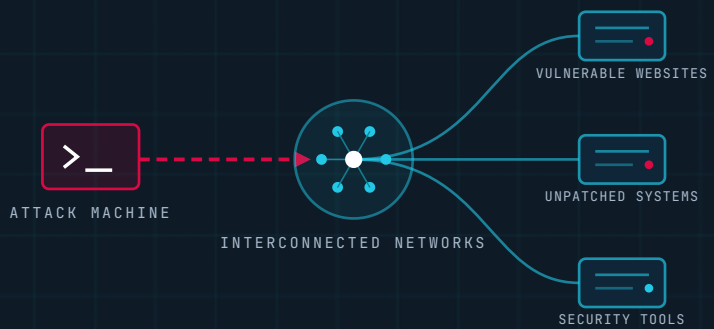
HACKING & SECURITY TOOLS

50%+

PRACTICAL

More than 50% of official
CEH training time is
dedicated to practical
activities.

CONTROLLED CYBER RANGE · AUTHORIZED PRACTICE



Depending on the module, the environment can include vulnerable websites, unpatched systems, interconnected networks, attack machines and security tools — so students perform authorized exercises rather than only watch demonstrations.

EXAMPLES OF PRACTICAL ACTIVITIES

- ✓ Reconnaissance & OSINT
- ✓ Network discovery
- ✓ Port & service scanning
- ✓ Enumeration
- ✓ Vulnerability assessment
- ✓ Authentication / password attacks
- ✓ System hacking exercises
- ✓ Network traffic analysis
- ✓ Web-server testing
- ✓ Web-application testing
- ✓ SQL injection exercises
- ✓ Wireless security
- ✓ Cloud security
- ✓ Container security
- ✓ Malware-related exercises
- ✓ AI-assisted ethical hacking

WHAT THE LAB JOURNEY FEELS LIKE

01
LEARN

02
PRACTICE

03
IDENTIFY

04
EXPLOIT SAFELY

05
UNDERSTAND

06
DEFEND

05 POWERED BY AI

AI in CEH v13

AI IS NOT SIMPLY ONE MODULE.

AI-assisted ethical hacking techniques are integrated throughout the CEH v13 learning journey.

THE 20-MODULE JOURNEY

DISCOVER → PROTECT

P1

DISCOVER

01 02 03
04

P2

**FIND THE
WEAKNESS**

05 06

P3

UNDERSTAND ATTACKS

07 08 09 10 11 12

P4

ATTACK SURFACES

13 14 15 16 17

P5

**MODERN
INFRASTRUCTURE**

18 19

P6

**PROTECT
INFORMATION**

20



AI-assisted ethical hacking

Integrated across appropriate areas of CEH v13



IN THE CURRICULUM

Across the learning journey

CEH v13 integrates AI-assisted ethical hacking across appropriate areas of the course rather than isolating it in a single module.



IN THE LABS

AI-assisted ethical hacking

Included among the practical activities carried out in the controlled cyber range, alongside reconnaissance, scanning, web, cloud and wireless exercises.



AS AN OUTCOME

Apply AI to cybersecurity

One of the eight core learning outcomes participants work towards by the end of the CEH v13 journey.

06 EXAMINATION & CERTIFICATION

CEH Examination & Certification

The CEH pathway separates knowledge validation from optional hands-on practical validation.

REQUIRED FOR CEH

EXAM CODE 312-50

CEH Knowledge Exam

125 Multiple-choice questions	4 HRS Exam duration
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Passing score: varies by exam form (EC-Council publishes a range)

TYPICAL AREAS TESTED

- Threats & attack vectors
- Reconnaissance / footprinting
- Scanning & enumeration
- Vulnerability analysis
- System hacking phases
- Malware, sniffing & social engineering
- DoS and session hijacking
- Web / SQL / wireless security
- Mobile, IoT/OT, cloud & cryptography
- Detection, prevention and countermeasures

PASS →
Certified Ethical Hacker (CEH)

OPTIONAL

HANDS-ON VALIDATION

CEH Practical

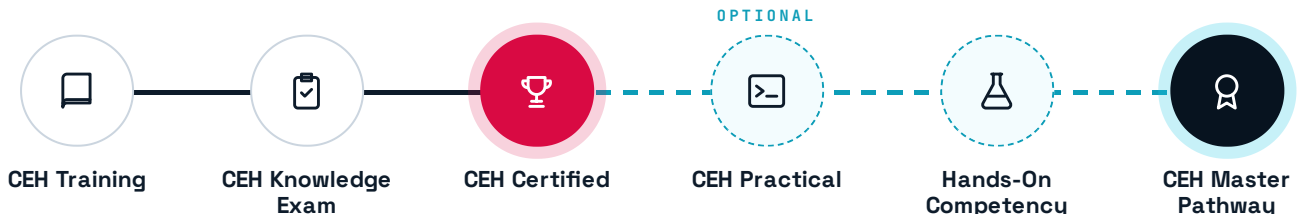
20 Real-world challenges	6 HRS Exam duration
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Format: Hands-on cyber range · **Environment:** EC-Council iLabs Cyber Range

PRACTICAL AREAS CAN INCLUDE

- Port and network scanning
- Vulnerability detection
- System attacks
- Session hijacking
- Web-server attacks
- Web-application attacks
- SQL injection
- Wireless security
- Protocol analysis
- Security testing tools

CEH + CEH PRACTICAL →
Supports progression to CEH Master



CEH Practical is optional.

Passing the CEH Knowledge Exam earns the CEH certification. The Knowledge Exam is the primary CEH certification exam; the Practical adds hands-on validation and is part of the CEH Master pathway.

07 LEARNING OUTCOMES

What will I actually learn?

BEFORE

“I understand cybersecurity.”



AFTER CEH v13

“I understand how attacks **actually** happen.”

By the end of the learning journey, participants should understand how attackers discover targets, identify weaknesses, exploit common attack surfaces in controlled environments and apply defensive countermeasures.

01



UNDERSTAND THE ATTACKER

Learn how attackers think and operate.

02



DISCOVER TARGETS

Collect intelligence and discover systems.

03



FIND VULNERABILITIES

Identify weaknesses before malicious attackers exploit them.

04



UNDERSTAND EXPLOITATION

Understand exploitation from an ethical hacker's perspective.

05



TEST DIFFERENT ATTACK SURFACES

Test major attack surfaces — from web and wireless to mobile, IoT/OT and cloud.

06



UNDERSTAND DEFENSE

Apply appropriate defensive countermeasures.

07



APPLY AI TO CYBERSECURITY

AI-assisted ethical hacking, integrated across the journey.

08



BUILD PRACTICAL ETHICAL HACKING SKILLS

Authorized exercises in a controlled cyber range.

ATTACKER MINDSET

DEFENSIVE CAPABILITY

08 CAREER PATHWAY

Where can CEH take you?

POSSIBLE ROLES

**Cybersecurity Analyst****SOC Analyst****Security Analyst****Information Security Analyst****Vulnerability Assessment Analyst****Security Administrator****Network Security Engineer****Cyber Defense Analyst****Cybersecurity Auditor****Information Security Auditor****Penetration Testing / Ethical Hacking roles****Cybersecurity Consultant**

A possible progression

FUNDAMENTALS → LEADERSHIP

01 IT / Networking Fundamentals

02 Cybersecurity Fundamentals

03 **CEH v13**

04 Junior Security / SOC / Vulnerability Role

05 Cybersecurity Analyst / Security Engineer

06 Choose Your Specialization

Ethical Hacking & Penetration Testing

SOC & Threat Detection

Incident Response

Cloud Security

Network Security

Digital Forensics

07 Senior Security / Consulting

08 **Security Architecture / Management**

Certification is one part of a cybersecurity career. Employers may also consider practical ability, IT fundamentals, experience and complementary technical skills.

YOUR NEXT STEP

READY TO START YOUR ETHICAL HACKING JOURNEY?

— Certified Ethical Hacker (CEH) v13 · Powered by AI



WEBSITE

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REGISTRATION INFORMATION

✉ Email

📅 Intake dates

📍 Venue / delivery mode

📋 Registration details



QR CODE